

Role Stress and Effectiveness in Horizontal Alliances

Horizontal arrangements are increasingly deployed in organizational networks, yet research has rarely examined the effectiveness of these alliances. The coalition of disparate corporate cultures yields appreciable levels of role stress for people in boundary-spanning positions. Dedicated assets and communication modality are factors that influence the level of role ambiguity and conflict. The authors implicate these facets of role stress as antecedents to four forms of effectiveness drawn from the competing values framework. The authors present alternative perspectives that examine the relationship between stress and performance. The received view frames role stressors as linear, negative antecedents to organizational outcomes. The authors contrast this perspective with theories that espouse triphasic, parabolic, and interactive influences of stressors on organizational outcomes. Data gathered with 218 managers of dual-branded retail oil outlets indicate that the relevance of these alternative perspectives is mitigated by the form of effectiveness pursued by the organization. The results support a linear relationship between role conflict and bargaining efforts, yet they also offer evidence of nonlinear influences of role ambiguity on contributions to sales, customer satisfaction, and competence. The study concludes with a discussion of relevance of the findings to the management of horizontal alliances and to interorganizational theory.

Marketing practice and thought have benefited substantially from the adoption of a long-term perspective in the analysis of exchange. Theory has been developed that addresses the domestication of markets (Arndt 1979) and the breadth of governance structures employed to manage long-term vertical exchange relationships (Heide 1994).

Increases in the magnitude of horizontal alliances have been concomitant with the rise in emphasis on long-term vertical exchange. Merger-related activities surpassed US\$2.30 trillion in 1998, and the top ten mergers alone exceeded US\$550 billion (Jackson 1999), yet empirical work has not kept pace with the rise in horizontal alliances. Bucklin and Sengupta (1993) examine success-related factors in comarketing pacts, and Johnson and Houston (2000) offer evidence of synergistic returns in horizontal joint ventures. Unfortunately, complementary work has not addressed interaction and performance in other horizontal arrangements.

Mergers and acquisitions are horizontal exchange relationships that dynamically change the organizational structure of multiple entities. Parties to these relationships seek scale economies in production and marketing, yet research suggests that many mergers have a negative influence on profits (Cartwright and Cooper 1995). The new organization must not only consolidate supply chain factors but also bring

together disparate corporate cultures. Nevertheless, marketing research has not examined the organizational challenges that emerge in the wake of a horizontal alliance. *Role ambiguity* emerges because of the paucity of relevant information as well as the level of complexity and change (Kahn et al. 1964). *Role conflict* similarly materializes as employees evaluate responsibilities in light of historical obligations and associations with new partners. Ambiguity and conflict jeopardize efforts to achieve objectives, but research has not examined stress in mergers and acquisitions.

The goal of this research is to examine role stress in horizontal alliances. Three objectives underlie our pursuit of this goal. The first objective is to examine influences of ambiguity and conflict on multiple facets of effectiveness. The competing values framework (Cameron and Quinn 1999) emphasizes the simultaneous pursuit of several dimensions of performance. Examinations of the determinants of a single facet of performance are ill equipped to evaluate the consequences of these antecedents to other channel outcomes (Kumar, Stern, and Achrol 1992). Focusing on a single element of performance constrains advancements in theory and limits researchers' ability to make policy recommendations (see Singh 2000). Our research, however, augments channel thought with an examination of role-based antecedents to multiple facets of effectiveness. Analysis of diverse facets of effectiveness informs managers about the multiple consequences of stress.

Our second objective is to examine alternative theoretical perspectives on the influences of role stress on organizational outcomes. Role stress is often implicated as a determinant of performance, yet prior research has not consistently supported this hypothesis (Tubre and Collins 2000). We present the linear negative influence of stress on performance (House and Rizzo 1972) as the *received view*. We contrast this perspective with theories that espouse triphasic (Selye 1950), parabolic (Yerkes and Dodson 1908),

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and interactive influences (Fried et al. 1998) of stressors on performance. We contribute to contemporary role research by offering evidence that the influences of role stress vary on the basis of the form of effectiveness. Moreover, managers of horizontal alliances can gain insight into the positive and negative influences of stress.

The third objective of our research is to identify organizational and communicative processes that influence the level of stress in alliances. Williamson (1985) underscores the importance of transaction-specific assets, yet the influence of these assets on stress has not been examined. Prior research offers insight into the content and flow of communication, but the medium employed for interaction has received less attention (Frazier 1999). Identification of these factors enables management to prepare for conflict and ambiguity. Furthermore, identification of these factors enables managers to discount some opportunities because of inevitably high levels of stress.

Our empirical context is the distribution system of two oil companies that operate through a horizontal alliance at the retail level. Mergers or acquisitions have recently been completed in a variety of retail settings (Zecher 1998), and the retail sector accentuates some inherent challenges associated with interfirm alliances. Retail executives who supervise mergers must monitor multiple aspects of integration that are deemed critical to the success of alliances. These integration issues include consolidation of financial systems; product delivery systems; pricing, promotion, and customer service policies; information systems; and human resources (Harvey, Price, and Lusch 1998). There is often a significant gap between the expected value of an acquisition and its actual performance, and the acquisition may lead to layoffs or divert attention away from core brands and markets (Weston and Chiu 1996). Brand managers should value research that addresses efforts to integrate divergent firms into a unified, effective enterprise. Nevertheless, research has rarely addressed postmerger integration (Kerin and Varaiya 1985; Kumar, Kerin, and Pereira 1991).

The article proceeds as follows: In the next section, we present our theoretical model and hypotheses. We then present the method and data collection procedures, along with the results. The final section outlines implications of the research for channel management and theory.

Model and Hypotheses

Role theory uses the metaphor of playacting to characterize the interaction between a focal person and a role sender (Kahn et al. 1964). The focal person encounters pressure because of expectations placed on him or her by role senders. For example, franchisees are endowed with role expectations from McDonald's upon investing in the franchised system.

Inherent to the nature of boundary-spanning positions is the likelihood that people occupying these roles will experience ambiguity and conflict. *Role ambiguity* refers to the "lack of clarity and predictability of the outcomes of one's behavior" (House and Rizzo 1972, p. 475). Interfirm theory and research on boundary-spanning occupations indicate that organizational factors and coping mechanisms have

important influences on role clarity (Singh 1998). Standard industry practice is to develop comprehensive operating systems that are at the retail managers' disposal (Keating 1991), yet retailers still encounter ambiguity. For example, retail managers experience ambiguity when retail traffic history is unavailable to assist in the scheduling of personnel.

Role conflict refers to the "degree of incongruity or incompatibility of expectations associated with the role" (House and Rizzo 1972, p. 475). House and Rizzo's analysis of conflict addresses incompatibilities among standards or values, time, resources, or capabilities; multiple role responsibilities; and various organizational inputs. Prior studies of boundary personnel recognize that each of these facets may influence conflict (Ford, Walker, and Churchill 1975; Singh, Verbeke, and Rhoads 1996). Retail managers encounter incompatibilities emanating from a variety of sources. For example, franchisees experience conflict when order quantities suggested by the franchisor are not consistent with the recommendations of local suppliers.

The Received View of Role Stress and Organizational Outcomes

Our theoretical model incorporates multiple perspectives on the relationship between stress and performance in horizontal alliances. The influences of ambiguity and conflict on performance have been the focus of considerable research, yet the findings do not support a strong, direct relationship (see Table 1). Consequently, researchers have proposed nonlinear (e.g., Schuler 1980) and interactive relationships (Fried et al. 1998) between stress and performance. Our model examines the independent influences of role stress on organizational outcomes and treats nonlinear and interactive effects as competing hypotheses. Treatment of conflict and ambiguity as independent factors is consistent with Kahn and colleagues' (1964) framework, yet it stands in contrast to several analysts' (e.g., Dubinsky et al. 1992) treatment of conflict as antecedent to ambiguity. Because the conditions leading to the sources of stress are similar, Kahn and colleagues (1964) address the independent effects of ambiguity and conflict and posit modest correlation among facets of stress. We incorporate this premise into our analysis.

We present the competing values framework (Cameron and Quinn 1999) as a means to contrast alternative stress theories and differentiate among forms of performance. The competing values framework reveals two major dimensions of effectiveness. The first dimension differentiates effectiveness on the basis of the extent to which adaptation and change are valued over predictability and stability. The second dimension contrasts an internal orientation that emphasizes unity and integration with an external orientation that focuses on differentiation and rivalry. Simultaneous consideration of the two dimensions reveals four perspectives on effectiveness. The *clan* approach reflects an internal emphasis with individual discretion. Its polar opposite, the *market* approach, refers to an externally oriented firm that values control. The *adhocracy* approach refers to an external orientation with value placed on flexibility. The *hierarchical* approach contrasts with adhocracy through an internal emphasis coupled with a desire for stability.

Organizations vary in the extent to which they emphasize one form of effectiveness over other facets. Our analy-

TABLE 1
Empirical Analyses of Satisfaction and Organizational Outcomes in Role Stress Research

Authors	Role Constructs ^a			Setting	Results
	Role Ambiguity Items	Role Conflict Items			
Donnelly and Ivancevich (1975)	3	3	86 sales representatives, 46 supervisors	Ambiguity is negatively associated with three facets of satisfaction.	
Churchill, Ford, and Walker (1976)	41: 21 company, 7 manager, 7 customer, 6 family	30: company, manager, customer, family	265 sales representatives	Ambiguity lowers job, coworker, company, and pay-based satisfaction. Conflict lowers supervisor and company-based satisfaction.	
Oliver and Brief (1977–78)	6 (Rizzo, House, and Lirtzmann 1970)	8 (Rizzo, House, and Lirtzmann 1970)	114 retail managers	Ambiguity and conflict lower satisfaction.	
Bagozzi (1978)	12 (Ford, Walker, and Churchill 1975)	15 (Kahn et al. 1964)	161 industrial sales representatives	Conflict and ambiguity are negatively associated with sales and job satisfaction. Stepwise regression suggests that only conflict lowers sales and satisfaction.	
Busch and Bush (1978)	5 (Donnelly and Ivancevich 1975)		78 industrial sales representatives, 39 women, 39 men	Ambiguity is negatively associated with performance. Ambiguity is negatively associated with customer (female representatives), work, and supervisor satisfaction.	
Dubinsky and Mattson (1979)	24: company, supervisors, customers	24: company, supervisors, customers	203 retail sales representatives	Conflict and ambiguity are negatively associated with performance and satisfaction.	
Teas, Wacker, and Hughes (1979)	6 (Rizzo, House, and Lirtzmann 1970)		107 sales representatives	Ambiguity lowers higher-order need fulfillment.	
Bagozzi (1980)	12 (Ford, Walker, and Churchill 1975)	15 (Kahn et al. 1964)	122 industrial sales representatives	Role ambiguity is negatively associated with sales and satisfaction in the structural equation model, but conflict is not related to either endogenous variable.	
Berkowitz (1980)	5 (Hamner and Tosi 1974)	15 (Secord and Backman 1970)	49 sales managers, 148 sales representatives	Ambiguity raises the sales-to-expense ratio for sales managers. Ambiguity and conflict lower job satisfaction.	
Kelly and Hise (1980)	5 (Lyons 1971)	8 (Rizzo, House, and Lirtzmann 1970)	198 brand managers	Ambiguity and conflict lower satisfaction.	
Kelly, Gable, and Hise (1981)	5 (Lyons 1971)	8 (Rizzo, House, and Lirtzmann 1970)	179 chain store managers	Ambiguity and conflict are unrelated to performance. Ambiguity is negatively related to job satisfaction.	

TABLE 1
Continued

Authors	Role Constructs ^a			Setting	Results
	Role Ambiguity Items	Role Conflict Items			
Behrman, Bigoness, and Perreault (1981)	19: 4 customer, 5 company, 5 family, 5 manager (Ford, Walker, and Churchill 1975)			193 industrial sales representatives	Family-based ambiguity raises performance. Manager- and customer-based ambiguity lower performance. Manager-based ambiguity lowers satisfaction.
Teas (1983)	(Rizzo, House, and Lirtzmann 1970) ^b	(Rizzo, House, and Lirtzmann 1970) ^b		116 sales managers	Conflict lowers satisfaction.
Behrman and Perreault (1984)	8 (Rizzo, House, and Lirtzmann 1970)	18 (Rizzo, House, and Lirtzmann 1970)		196 sales representatives	Ambiguity lowers and conflict raises performance. Ambiguity and conflict lower job satisfaction.
Kohli (1985)	6 (Rizzo, House, and Lirtzmann 1970)			114 sales managers	Ambiguity lowers satisfaction.
Chonko, Howell, and Bellenger (1986)	36: 3 family, 12 job, 7 company, 7 supervisor, 7 customer	30: 4 family, 4 job, 11 supervisor, 6 customer, 5 self		121 industrial sales representatives	Congruence of evaluation and compensation is lowered through job-related ambiguity, but it is raised through company and supervisor ambiguity.
Fry et al. (1986)	8 (Rizzo, House, and Lirtzmann 1970)	7 (Rizzo, House, and Lirtzmann 1970)		216 sales representatives	Ambiguity lowers job, company, and customer satisfaction. Conflict lowers all INDSALES facets except customer satisfaction.
Hampton, Dubinsky, and Skinner (1986)	6 (Rizzo, House, and Lirtzmann 1970)	8 (Rizzo, House, and Lirtzmann 1970)		116 retail sales representatives	Ambiguity and conflict are unrelated to performance. Conflict lowers overall satisfaction.
Michaels, Day, and Joachimsthaler (1987)	10 (Rizzo, House, and Lirtzmann 1970)	13 (Rizzo, House, and Lirtzmann 1970)		952 industrial buyers	Ambiguity lowers and conflict raises performance. Both role factors lower satisfaction.
Yammarino and Dubinsky (1990)	6 (Rizzo, House, and Lirtzmann 1970)	5 (Dansereau, Graen, and Hage 1975)		109 retailers, 116 insurance agents	Conflict and ambiguity are negatively associated with performance.
Lusch and Jaworski (1991)	4 (Kahn et al. 1964)	4 (Kahn et al. 1964)		182 retail store managers	Role stress lowers performance.
Singh and Rhoads (1991)	45: 9 company, 9 boss, 8 customer, 6 ethical, 4 other managers, 5 coworkers, 4 family			472 SMEs: sales and marketing executives	All seven dimensions are negatively associated with performance and satisfaction.

TABLE 1
Continued

Authors	Role Constructs^a			Setting	Results
	Role Ambiguity Items	Role Conflict Items			
Dubinsky et al. (1992); Singh (1995)	9 (Rizzo, House, and Lirtzmann 1970)	11 (Rizzo, House, and Lirtzmann 1970)		Sales representatives: U.S., 218; Japan, 220; Korea, 156	Ambiguity lowers performance; conflict has a positive, direct effect on performance, but the total effect is nonsignificant. Conflict and ambiguity (Japan only) lower satisfaction.
Singh (1993)	45: 9 company, 9 boss, 8 customer, 6 ethical, 4 other manager, 5 coworkers, 4 family (Singh and Rhoads 1991)			472 SMEs: marketing executives, 254 IS marketing and support representatives	SME: Customer-based ambiguity lowers performance, company and supervisor ambiguity lower satisfaction, and family-based ambiguity raises satisfaction. IS: company, customer, and other ambiguity lower performance; supervisor-based ambiguity lowers satisfaction.
Singh, Goolsby, and Rhoads (1994)	3 (Rizzo, House, and Lirtzmann 1970)	3 (Rizzo, House, and Lirtzmann 1970)		377 customer service representatives	Direct effects of conflict and ambiguity on performance are not significant. Burnout mediates the effects of stress on performance and satisfaction.
Singh, Verbeke, and Rhoads (1996)	3 (Rizzo, House, and Lirtzmann 1970)	3 (Rizzo, House, and Lirtzmann 1970)		188 boundary spanners	Ambiguity lowers performance, and the effect is moderated by organizational archetype. Ambiguity lowers satisfaction. Conflict lowers satisfaction for the affective archetype.
Hartline and Ferrell (1996)	17 (Chonko, Howell, and Bellenger 1986)	12 (Chonko, Howell, and Bellenger 1986)		797 hotel employees	Conflict lowers self-efficacy. Ambiguity lowers self-efficacy and satisfaction.
Singh (1998)	7 (Rizzo, House, and Lirtzmann 1970)	8 (Rizzo, House, and Lirtzmann 1970)		703 marketing executives and representatives	Ambiguity and conflict lower performance. Conflict lowers satisfaction.
Singh (2000)	9 company, 8 customer (Singh and Rhoads 1991)	2 intersender, 4 resource demands (House 1980)		182 customer support representatives, 177 bill collectors	In both samples, customer-based ambiguity lowers quality and productivity, and resource-based conflict lowers productivity.

^aNames in parentheses correspond with the sources of the measures.

^bThe number of items was not reported.

Notes: Churchill, Ford, and Walker's (1974) INDSALES assesses satisfaction with the job, fellow workers, supervision, company, pay, promotion potential, and customers. SMEs = Small and medium-sized enterprises. IS = information systems.

sis focuses on multiple facets of effectiveness pursued at the retail level. Table 2 summarizes research on the relationship between effectiveness and stress and categorizes operationalizations within the competing values framework. Most recent research acknowledges multiple facets of effectiveness, yet the empirical analyses develop composite measures that are drawn from multiple, competing forms of effectiveness (see Singh 2000). For example, Michaels, Day, and Joachimsthaler (1987) describe six elements of performance, yet their analysis solely addresses the composite measure. Composite measures provide insight into overall effectiveness, yet the influence on independent facets of performance is not established. In contrast, our model examines influences of stress on each mode of effectiveness. Consider first the influence of stress on the clan perspective.

Clan-based effectiveness and role stress. The clan-based approach emphasizes the shared goals and values of team members in conjunction with participative decision making. The primary tasks of managers in this approach are to empower employees and foster their commitment and loyalty to the organization (Cameron and Quinn 1999). Our treatment of clan-based effectiveness examines the competence of the retailer. *Competence* reflects the experience, product knowledge, and capabilities acquired at the retail site (see Kumar, Stern, and Achrol 1992). Domino's Pizza, for example, attributes its success to the experience and skills acquired by franchisees (Boroian and Boroian 1997).

Research with organizational teams (Orsburn et al. 1990) suggests that clearly defined role responsibilities influence efforts to achieve confidence. The human resources within a team are unlikely to be channeled toward common objectives unless the responsibilities of the managers are well defined. Clear expectations enable retailers to gain confidence in the abilities of their trading partners. In contrast, poorly defined roles preclude personnel from recognizing expectations and foster lower perceptions of competence (Dubinsky and Mattson 1979; Singh 1998).

Role conflict should also influence competence. Research by Cohen (1980) suggests that role stress affects the availability of cognitive resources. As more resources are dedicated to coping with the stress, fewer resources are devoted to performing job responsibilities. Therefore, Dubinsky and Mattson (1979) report a negative relationship between conflict and retail salespeople's perceptions of competence. As stress-related demands increase, local managers are less likely to recognize factors critical to success and become uncertain about their job-related abilities. Inconsistencies in the messages sent by role partners should cloud managers' evaluations of their capabilities.

H_{1a}: The manager's role ambiguity is negatively associated with managerial competency.

H_{2a}: The manager's role conflict is negatively associated with managerial competency.

Adhocracy-based effectiveness and role stress. An adhocracy parallels a clan in its emphasis on flexibility and discretion, yet an adhocracy places greater emphasis on external positioning and differentiation. Our adhocracy-based analysis addresses local efforts to enhance customer satisfaction. Copy Mat photocopy systems offer evidence of the value

derived from attending to customer satisfaction (Boroian and Boroian 1997). Although operating in a commodity business that has a poor reputation for quality, the company succeeds through managerial efforts to raise customer satisfaction.

Substantial research has addressed the relationship between role stress and boundary spanners' satisfaction with customers, yet few studies have considered the influence of stress on efforts to enhance customer satisfaction. When sufficient knowledge is not provided, boundary spanners lack the information necessary to attend to the needs of buyers. Singh (1993) reports that performance wanes in the presence of customer-based role ambiguity encountered by sales and customer service agents. Role ambiguity among retailers should have a strong influence on the level of customer satisfaction. When retailers perceive that obligations associated with job performance are unclear, they should experience greater difficulty in their efforts to meet customer expectations.

Role conflict should have a similar influence on efforts to enhance customer satisfaction. Pfeffer and Salancik (1978) maintain that competing demands inhibit people from achieving objectives. Retailers faced with the demands of multiple parties may perform tasks that have countervailing influences on customer outcomes. Conflicting demands may also lead retailers to prioritize obligations to role partners and downplay customer needs. In either case, the likelihood of addressing the demands of external constituents is likely to suffer.

H_{3a}: The manager's role ambiguity is negatively associated with managerial efforts to enhance customer satisfaction.

H_{4a}: The manager's role conflict is negatively associated with managerial efforts to enhance customer satisfaction.

Market-based effectiveness and role stress. Market-based organizations focus on relationships external to the firm, yet they simultaneously value stability. Market-based effectiveness emphasizes performance compared with competition and is embodied in contribution to sales (see Kumar, Stern, and Achrol 1992). This construct reflects the revenues generated by a location as compared with other dealers. For example, to varying degrees, managers engage in local marketing activities that are designed to fuel sales.

A fundamental premise of the rational goal approach is the notion that organizations are collectivities oriented to specific goals. Ambiguity and conflict constrain efforts to achieve these goals. Consistent with this perspective, Bagozzi (1980) reports a negative influence of role ambiguity on sales performance. As role responsibilities become less clear, retail managers must dedicate more cognitive resources to identify role expectations. Because cognitive resources are limited, resources allocated to clarifying responsibilities cannot be dedicated to pragmatic, goal-based action (Cohen 1980). Consequently, managers encountering unclear expectations should contribute less to the firm's revenue stream. An increase in the level of conflict similarly requires managers to dedicate more energy to sorting out the demands of multiple stakeholders. As more resources are allocated to sorting out role obligations, contributions to performance should falter.

H_{5a}: The manager's role ambiguity is negatively associated with managerial contributions to sales.

H_{6a}: The manager's role conflict is negatively associated with managerial contributions to sales.

TABLE 2
Operationalizations of Satisfaction and Organizational Outcomes in Role Stress Research

Authors	Competing Values Framework: Conceptual Domain of the Performance Items					
	Satisfaction Items ^a	Performance Items ^a	Human Relations	Open Systems	Rational Goal	Internal Process
Donnelly and Ivancevich (1975)	6					
Churchill, Ford, and Walker (1976)	95 (Churchill, Ford, and Walker 1974) ^b					
Oliver and Brief (1977–78)	18 (Brayfield and Rothe 1951)					
Bagozzi (1978)	8 (Pruden and Reese 1972)	1			1 Sales	
Busch and Bush (1978)		1 (Pruden and Reese 1972) ^c				
Dubinsky and Mattson (1979)		1 (Pruden and Reese 1972) ^c				
Teas, Wacker, and Hughes (1979)	13 (Porter and Lawler 1968)					
Bagozzi (1980)	8 (Pruden and Reese 1972)	1			1 Sales	
Berkowitz (1980)	9 (Ivancevich and Donnelly 1974)	5 ^c			1 Sales 1 Turnover	1 Net profit 1 Sales/expense ratio
Kelly and Hise (1980)	9 (Ivancevich and Donnelly 1974)					
Kelly, Gable, and Hise (1981)	9 (Ivancevich and Donnelly 1974)	5			1 Return on assets, 1 Volume/ 1 Volume/square foot, 1 Contribution, 1 Contribution/square foot	
Behrman, Bigoness, and Perreault (1981)	95 (Churchill, Ford, and Walker 1974) ^b	5	1 Technical knowledge 1 Providing information 1 Sales presentations		1 Meeting sales objectives	1 Controlling expense

TABLE 2
Continued

Competing Values Framework: Conceptual Domain of the Performance Items						
Authors	Satisfaction Items ^a	Performance Items ^a	Human Relations	Open Systems	Rational Goal	Internal Process
Teas (1983)	36 (Smith, Kendall, and Hulin 1969)					
Behrman and Perreault (1984)	95 (Churchill et al. 1974) ^b	31 (Behrman and Perreault 1982)	6 Technical knowledge 5 Providing information 6 Sales presentations		7 Quantity and quality sales objectives	7 Controlling expense
Kohli (1985)	13 (Porter and Lawler 1968)					
Chonko, Howell, and Bellenger (1986)		11 ^d	1 Ability to reach quota 1 Company knowledge 1 Customer knowledge 1 Product knowledge 1 Competitor knowledge	1 Customer relations	1 Sales volume 1 Percent increase in salary	1 Expense accounts 1 Time management 1 Planning
Fry et al. (1986)	95 (Churchill, Ford, and Walker 1974) ^b					
Hampton, Dubinsky, and Skinner (1986)	8 (Hackman and Oldham 1975)	1 (Pruden and Reese 1972) ^c				
Michaels, Day, and Joachimsthaler (1987)	18 (Smith, Kendall, and Hulin 1969)	6 (Reck 1978)	1 Departmental relations 1 Workload	1 Vendor relations	1 Price variances	1 Cost savings 1 Negotiations
Yammarino and Dubinsky (1990)	15 ^e (Dansereau, Graen, and Hage 1975)	5 ^e (Dansereau et al. 1982); 10 (retail), 12 (insurance)	1 Product knowledge 1 Resourcefulness 5 Satisfaction with performance			1 Profitability
Lusch and Jaworski (1991)		31 ^{c, e}	22 Competence 8 Satisfaction with performance			
Singh and Rhoads (1991)	26 (Churchill, Ford, and Walker 1974) ^b	6 ^e	1 Ability 1 Product knowledge 1 Performance potential	1 Customer relations	1 Quantity of sales	1 Time management
Dubinsky et al. (1992); Singh (1995)	5 (Hackman and Oldham 1975)	10 ^e (Yammarino and Dubinsky 1990)				

TABLE 2
Continued

Authors	Competing Values Framework: Conceptual Domain of the Performance Items					
	Satisfaction Items ^a	Performance Items ^a	Human Relations	Open Systems	Rational Goal	Internal Process
Singh (1993)	26 (Churchill, Ford, and Walker 1974) ^b	6 (Dubinsky and Mattson 1979)	1 Ability 1 Product knowledge 1 Performance potential	1 Customer relations	1 Quantity of sales	1 Time management
Singh, Goolsby, and Rhoads (1994)	26 (Churchill, Ford, and Walker 1974) ^b	6 (Dubinsky and Mattson 1979)	1 Ability 1 Product knowledge 1 Performance potential	1 Customer relations	1 Quantity of sales	1 Time management
Singh, Verbeke, and Rhoads (1996)	6 (Ironson et al. 1989)	6 (Dubinsky and Mattson 1979)	1 Ability 1 Product knowledge 1 Performance potential	1 Customer relations	1 Quantity of sales	1 Time management
Hartline and Ferrell (1996)	8 (Brown and Peterson 1993)	8 (Jones 1986)	8 Self-efficacy			
Singh (1998)	26 (Churchill, Ford, and Walker 1974) ^b	6 (Dubinsky and Mattson 1979)	1 Ability 1 Product knowledge 1 Performance potential	1 Customer relations	1 Quantity of sales	1 Time management
Singh (2000)		17 (Hartline and Ferrell 1993); ^f 7 developed		3 Trust, 5 promptness, 5 reliability, 4 attention	4 Contact outputs	3 Backroom work

^aNames in parentheses correspond to the sources of the measures.

^bChurchill, Ford, and Walker's (1974) INDSALES assesses satisfaction with the job, fellow workers, supervision, company, pay, promotion potential, and customers.

^cA single-item, global measure of performance was incorporated into this study.

^dIn this study, the effectiveness measure addressed the congruence between the sales representatives' increase in salary and the other performance factors.

^eAll items are not reported.

^fHartline and Ferrell's (1996) measures address quality, the new measures address productivity.

Hierarchy-based effectiveness and role stress. Interfirm relationships with a hierarchical orientation parallel market-based organizations in their emphasis on control, yet their internal focus is consistent with clan-based organizations. The simultaneous attention to stability and internal operations is reflected in values that center on efficiency and smooth-flowing production. Berkowitz's (1980) analysis of industrial sales territories is one of the few studies that links role clarity to sales-to-expense ratios. Our investigation of efficiency-based factors examines the degree to which negotiation between firms reflects *coordinated bargaining efforts*, which address the extent to which negotiation between the transacting parties is systematic and effective (Milgrom and Roberts 1991). For example, retailers negotiate with suppliers over order quantities, deliveries, training, and staffing.

Ambiguity and conflict are likely to inhibit efforts to coordinate bargaining sessions. When role obligations are poorly understood or underspecified, trading partners are inclined to use bargaining sessions as vehicles for clarifying responsibilities. In addition, managers experiencing high levels of ambiguity are unlikely to focus on relevant factors during negotiation periods. Conflict should similarly lead to lengthy negotiation sessions, but the motivations for negotiating are different. Retail managers who are cognizant of high conflict may view bargaining sessions as opportunities to reconcile disputes, and recognition of points of contention may lead them to stake out negotiation stances that are unacceptable to their trading partners. Conflict may also lead trading partners to avoid interaction and interact less openly when bargaining commences (see Michaels, Day, and Joachimsthaler 1987). Consequently, ambiguity and conflict require the dedication of additional resources to the bargaining process.

H_{7a}: The manager's role ambiguity is negatively associated with coordinated bargaining efforts.

H_{8a}: The manager's role conflict is negatively associated with coordinated bargaining efforts.

Alternative Perspectives on Role Stress and Organizational Outcomes

General adaptation syndrome (G-A-S). Many studies have examined relationships between role stress and effectiveness, yet the results do not unequivocally support the hypotheses. For example, analyses in retail (e.g., Kelly, Gable, and Hise 1981) and other contexts (Singh, Verbeke, and Rhoades 1996) fail to support the negative influences of stress on performance. Research in biology and psychology suggests that responses to stress may not be linear. Yerkes and Dodson's (1908) laboratory experiments illustrate a nonlinear relationship among habit formation, performance, and stressors. Their research indicates that intermediate levels of stimuli outperform weak or strong stimuli when tasks are challenging. Thus, stress is viewed as having a parabolic relationship with performance. Activation theory (Scott 1966) similarly maintains that low levels of stress constrain performance because of minimal stimulation. High levels of stress yield similar performance because of overstimulation

and disorganization of responses. Intermediate levels of stressors, however, provide appropriate stimulation to yield desirable outcomes.

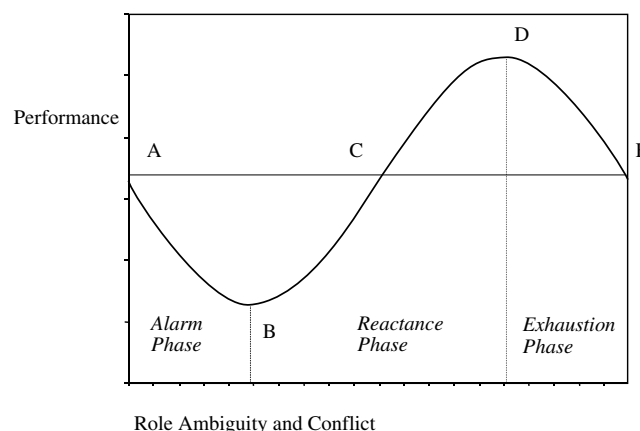
Marketing researchers (e.g., Singh 1998) have incorporated logic from habit formation and activation theory, but their results do not corroborate these perspectives. Selye (1950) offers a contrasting perspective in his presentation of the G-A-S as a three-phase model of reactions to stressors (see Figure 1). The alarm phase (A to B) is characterized by increasingly lower levels of performance. For example, biological research characterizes body weight loss as a symptom of this phase. During the reactance stage (B to D), performance factors increase and resistance to stress increases. Selye (1974) uses the term "eustress" to describe positive consequences of stress. Eustress is accompanied by coping behaviors that enable people to overcome stress and accomplish tasks that are considered worthwhile. Beyond some threshold, however, the exhaustion phase (D to E) is observed, in which reactions are similar to those in the initial phase.

The G-A-S perspective was developed from laboratory observations of within-subject responses to increasing levels of stimuli. The experimental results do not evince constant thresholds of responses to stress but reflect consistent patterns of results across subjects (Selye 1974). The thresholds (B and E) vary across subjects, but the triphasic response pattern should be consistent.

The periodicity outlined in G-A-S augments the hypotheses examined in prior research. The linear effects offered in many studies are embedded into the initial and final phases of the function (arcs A to B and D to E, respectively, in Figure 1). The function also incorporates logic from Yerkes and Dodson's (1908) research on habit formation to account for nonlinear influences of stress. This relationship is incorporated into the G-A-S function by the arcs from points C to E in Figure 1.

Our investigation of G-A-S effects examines the discrete influence of stress on four performance factors that are

FIGURE 1
General Adaptation Syndrome Effects of Stress on Performance



drawn from the competing values framework (Cameron and Quinn 1999). Interorganizational systems that value flexibility maintain that people should be granted the opportunity to implement their own styles to achieve objectives. This perspective emphasizes an internal orientation and personal flexibility and is embodied in an organization's efforts to nurture competence. Low levels of ambiguity and conflict provide limited opportunity to hone capabilities and yield lower levels of competency. At high levels, stress factors are overwhelming and similarly constrain competence. In contrast, intermediate levels of ambiguity and conflict pose challenges to boundary spanners, and accomplishments achieved in this context heighten competence.

H_{1b}: The manager's role ambiguity has a nonlinear, triphasic relationship with managerial competency.

H_{2b}: The manager's role conflict has a nonlinear, triphasic relationship with managerial competency.

Values related to the open systems framework underscore a firm's simultaneous emphasis on external positioning, flexibility, and individuality. When ambiguity and conflict are present in trivial amounts, managers have little incentive or opportunity to implement strategies that enhance customer satisfaction. As ambiguity and conflict increase, coping skills are developed that enable managers to initiate action that raises satisfaction. Nevertheless, the positive influence of stress operates within a limited spectrum. Excessive levels of ambiguity preclude managers from identifying action that is relevant to customer satisfaction. Extreme conflict requires the dedication of resources to evaluate and reconcile interrole disagreements. As the resources dedicated to reconciliation of responsibilities mount, less effort is channeled toward customer satisfaction.

H_{3b}: The manager's role ambiguity has a nonlinear, triphasic relationship with managerial efforts to enhance customer satisfaction.

H_{4b}: The manager's role conflict has a nonlinear, triphasic relationship with managerial efforts to enhance customer satisfaction.

The triphasic influence of stress outlined in the G-A-S should also be germane to contributions to sales. The rational goal model emphasizes the pursuit of productivity through a strong emphasis on external positioning and control. At low levels of ambiguity and conflict, retailers are not motivated to engage in activity to raise revenues. As clarity begins to wane and conflicts intensify, the retailer is likely to feel challenged and dedicates more energy to enhancing sales. At extreme levels, however, unclear expectations and inconsistencies in requirements preclude the manager from recognizing and engaging in efforts that enhance contributions to sales.

H_{5b}: The manager's role ambiguity has a nonlinear, triphasic relationship with managerial contributions to sales.

H_{6b}: The manager's role conflict has a nonlinear, triphasic relationship with managerial contributions to sales.

The G-A-S effects should also be germane to bargaining efforts. Initial observations of unclear guidelines and conflicting expectations warrant consideration in bargaining sessions. As ambiguity and conflict increase, managers develop skills that enable them to function efficiently in

interactions with brand management. At excessive levels, uncertainty and competing expectations void any gains associated with coping skills. Trading sessions become opportunities to clarify obligations and prioritize conflicting demands, and bargaining sessions are prolonged.

H_{7b}: The manager's role ambiguity has a nonlinear, triphasic relationship with coordinated bargaining efforts.

H_{8b}: The manager's role conflict has a nonlinear, triphasic relationship with coordinated bargaining efforts.

Interaction effects. Fried and colleagues (1998) offer a perspective on the influences of stress that stands in contrast to those outlined in the G-A-S. According to their approach, each facet of stress reduces a person's capacity to control the work setting. As multiple dimensions of stress materialize, the person's ability to maintain control over the work setting diminishes. Although people can control single components of stress, the simultaneous presence of ambiguity and conflict inhibits the ability to exercise control, which results in lower performance.

We propose that these interactive influences of ambiguity and conflict are relevant to organizational values that emphasize stability and control. Therefore, the pursuit of market-based objectives such as contributions to sales should be impaired by the simultaneous presence of ambiguity and conflict. When responsibilities are poorly defined, consistent role obligations emanating from multiple partners enable local retailers to attain acceptable sales levels. In contrast, when retailers are confronted with inconsistent demands from multiple trading partners, clearly delineated responsibilities enable retail managers to garner high contributions to sales. The recognition that roles are poorly specified and multiple partners are placing inconsistent demands, however, leads retail managers to generate lower contributions to sales.

The interactive influences of role stressors should also affect bargaining efforts. Ambiguities in obligations may not lead to higher bargaining expenditures when role expectations are compatible, yet well-defined roles may identify relevant bargaining issues despite substantial conflict. High conflict exacerbates role demands, and low clarity complicates efforts to prioritize expectations. Thus, effort is focused on addressing multiple demands throughout negotiations.

H_{5c}: The manager's role ambiguity and conflict have an interactive influence on contributions to sales. Managerial contributions to sales decrease as role ambiguity and conflict increase simultaneously.

H_{7c}: The manager's role ambiguity and conflict have an interactive influence on coordinated bargaining efforts. The degree to which bargaining efforts are coordinated declines as role ambiguity and conflict increase simultaneously.

Antecedents to Role Stress

Our model addresses economic factors and communication processes that influence the level of stress operating in alliances. Consider first how economic factors influence role stress.

Economic antecedents to role stress. Transaction cost analysis is a theory that underscores the importance of the

nature of investments made in a relationship (Williamson 1985). To varying degrees, assets dedicated to a relationship have limited value in other settings. For example, operating procedures for an online cash register system have limited value in other retail networks. Transaction cost theory refers to these investments as *transaction-specific assets*. Specific assets take on importance because managers must ensure that these investments are employed productively.

A manager's commitment to a retail distribution system involves several forms of capital that have limited value outside of the network. For example, hoteliers become well acquainted with the chain's reservation and operating systems (Shook and Shook 1993), and the nuances of the system cannot be redeployed in other hotel systems. Investments in learning the operating system of a retail network are investments in the role outlined by the role sender. Horizontal alliances complicate specific assets because managers enter the new contract with a history of dedicated investments. Managers carry over expectations based on their knowledge of the operating system developed for the old branded concept. These investments augment the expectations on the local managers and should raise conflict. Similarly, obligations endemic to the old system should escalate ambiguity for retailers. Investments in understanding the prior operating system complicate the managerial task to the extent that current policies are inconsistent with prior obligations. Managers more familiar with the old system should experience less clarity about expectations in the dual-branded system.

H₉: Transaction-specific investments in the former operating systems are positively associated with managerial role ambiguity.

H₁₀: Transaction-specific investments in the former operating systems are positively associated with managerial role conflict.

Investments in the current operating system also have implications for role stress. Agency contracts such as franchises are designed to bring the best possible outcome for the principal by aligning the incentives of both parties to the contract (Bergen, Dutta, and Walker 1992). As the agents dedicate more effort to learning the system, they are likely to recognize facets of the system that favor the principal. As a result of this process, they experience higher levels of conflict in their relationships with network management. Their investment should have a contrasting effect on ambiguity. Efforts to learn the operating system should clarify the principal's expectations.

H₁₁: Transaction-specific investments in the current operating systems are negatively associated with managerial role ambiguity.

H₁₂: Transaction-specific investments in the current operating systems are positively associated with managerial role conflict.

Communication-based antecedents to role stress. The mode of communication should have a strong influence on retail managers' stress. The medium used to transmit information has been operationalized in a variety of ways, including personal/impersonal and formal/informal means. Huber and Daft (1987) present a hierarchy of communica-

tion modes that ranges from face-to-face contact to less personal communiqués such as form letters. As the vehicle becomes more personal and multidimensional, greater opportunity is presented to interpret and transmit information. For example, face-to-face communication provides the opportunity to send verbal and nonverbal messages, yet form letters are more sterile means of communication. Employees prefer personal communication because it provides opportunities for dialogue that is more difficult to achieve in other modes (Larkin and Larkin 1996). Personal communication underscores the social link between the values of the employee and the firm, and conflict emerges when this link breaks down (Strebel 1996). Dialogue enables employees to express contingencies associated with multiple role obligations, and resolutions can be designed that meet each party's needs without raising conflict. Hallowell (1999) uses similar logic in his description of the human moment in interpersonal communication. The human moment involves occupying the same physical space and dedicating physical and emotional attention to interaction. Electronic media offer efficiencies in transmissions of sterile information, but personal cues cannot be transmitted as effectively in modes such as fax or e-mail. Body language, intonation, and emotion are constrained in less personal communication. Group gatherings and written correspondences are less capable of transmitting the meaning embodied in nonverbal communication. The meaning underlying words is muddled as the communication becomes less personal and ambiguity becomes more pronounced.

H₁₃: Personal modes of communications are negatively associated with managerial role ambiguity, yet group and impersonal modes are neither positively nor negatively associated with role ambiguity.

H₁₄: Personal modes of communications are negatively associated with managerial role conflict, yet group and impersonal modes are neither positively nor negatively associated with role conflict.

Method

Empirical Context

The Norwegian distribution system of two oil refiners served as the setting for our study. The refiners recently elected to merge their operations at the retail level. The stations adopted dual-branded trademarks and signs, and their shops were redesigned to reflect uniformity in the product setting and offerings. The retail setting provided a consistent image for consumers, yet the retail operating system brought together two corporate systems. The Norwegian state-owned company became a retailer in 1986 through the purchase of the retail network of a global refiner. Before the merger, this company had limited access to oil retailing. The global partner began Norwegian operations in 1930 and established a highly bureaucratic system with extensive reporting lines.

An internal proprietary study suggested that stress increased after the merger. The proprietary study, collected at the same time as the focal survey, revealed that managers believed that role responsibilities within the system were unclear and managerial decision making was performed on

a day-to-day basis. Similarly, area supervisors indicated that they lacked the responsibility to perform their jobs properly. Managers reported that information related to new programs was insufficient, and no area supervisor could describe the strategy for the market or for the dual-branded concept. Unclear internal responsibilities, ad hoc decision making, and limited empowerment of supervisors fueled conflict in the system, and insufficient information in the system exacerbated role ambiguity.¹

The method for analysis was a between-subjects design, and data were collected with retailers and the refiners' area managers. Much of the theory pertaining to responses to stress has been developed in laboratory settings that employ within-subjects designs. The laboratory setting provides an opportunity to decrease the variance in response to stress (Carroll and Nelson 1993), but a controlled within-subjects design is rarely feasible in applied settings. The effects are weaker in a between-subjects design, yet their observation suggests that the effects are appreciable (Grice and Hunter 1964).

Sampling Procedure and Data Collection

The data were collected through surveys sent to each retail manager in the network. Respondents were informed by telephone that they were about to receive the survey. The survey was accompanied by supporting letters from the retail union, the refiners, and the principal researcher. The response rate was 55.9% (218 of 390 responses). Among the responding firms, 89 had been associated with the state-owned network, and 129 had been associated with the global refiner. The mean level of managerial experience was 16.4 years, and all managers had at least one year of experience before the merger. Analysis of several variables (e.g., annual sales) and the focal constructs indicated no significant differences between early and late responses (Armstrong and Overton 1977). This analysis suggests that nonresponse bias is not an issue, but a stronger test would have been to contact nonrespondents.

Measure Development

Measure development followed the guidelines outlined by Gerbing and Anderson (1988). The survey instrument was developed with retail management. The director of the retail network, the director of the retail managers' union, and an industry consultant provided insight into the proper wording of the measures. In addition, pretests with four managers offered us an opportunity to make semantic changes in variables and make the study more meaningful to the respondents.

Although additional respondents enable researchers to control for informant bias (Kumar, Stern, and Anderson 1993), the pilot study indicated that additional respondents were not available at the retail level. Nevertheless, we

employed a matching dyad approach using data gathered with the refiners' area supervisors. Given that supervisors oversee four to ten retail outlets, it was infeasible to obtain reports for each outlet. We obtained 71 reports, which reflected a 95% response rate from supervisors. The manager and supervisory surveys used parallel wording. We employed coefficient alpha, item-to-total correlation, and exploratory factor analysis to purify the measures. We eliminated items that did not load properly in either factor analytic procedure. Eigenvalues and scree tests illustrated that one factor was sufficient to represent the items within each conceptual domain.

Measures

Transaction-specific assets. The dealer's investments in the current and prior operating systems were analyzed by separate sets of measures drawn from Anderson's (1985) company-specific asset scale. Four seven-point Likert-type items assessed the investment dedicated to learning the retail operating system.

Communication modality. We assessed this construct using formative measures of the frequency of usage of communication channels (Mohr, Fisher, and Nevin 1996). Managers interact with the refiner through face-to-face contact, telephone, letters, Internet, union meetings, seminars, and local retailer meetings. The refiner uses these modes but also communicates through advertisements, sales literature, and newsletters. We grouped the measures into three categories: personal (face-to-face messages and telephone calls), group (union meetings, local managers' meetings, and seminars), and impersonal interactions (letters and Internet contacts, refiner promotions, sales, and newsletters).

Role stress. The role stress scales were based on the seven-point measures developed by Rizzo, House, and Lirtzmann (1970). The conflict and ambiguity measures consisted of six and seven items, respectively.

Competence. This construct was analyzed by two seven-point items that addressed the knowledge and skills acquired by the local retail personnel. The scale was adapted from Kumar, Stern, and Achrol (1992).

Customer satisfaction. The satisfaction scale was based on Kumar, Stern, and Achrol's (1992) customer satisfaction scale. Three seven-point Likert items assessed efforts to achieve customer satisfaction.

Contributions to sales. This construct was assessed through three seven-point Likert items derived from Kumar, Stern, and Achrol (1992). The items addressed sales' contributions compared with competitors' and other retailers' sales.

Coordinated bargaining efforts. This construct examined the extent to which negotiation sessions were systematic and effective. Two seven-point items drawn from Nygaard (1992) were employed.

Construct Validity

We employed Anderson and Gerbing's (1988) two-phase approach to assess the model. We assessed the first phase, confirmatory factor analysis of the reflective measures,

¹After the data-collection period, the refiners implemented two systems to alleviate stress associated with the transition to the dual brand. The refiners established a program designed to guarantee consistency in role expectations and reporting responsibilities throughout the system, and they developed a comprehensive online operating system intended to clarify managerial role expectations.

using EQS. Initial analysis ($\chi^2(406) = 635.546$, $p < .05$; comparative fit index [CFI] = .921) indicated that elimination of some items would enhance the fit indices. Nevertheless, pursuit of optimal fit can limit the conceptual domain. Standardized residuals and Lagrange multipliers indicated significant cross-loadings for one conflict item. Removal of the item (see the Appendix) offered a stronger representation of the data ($\chi^2(377) = 589.979$, $p < .05$; CFI = .926). We assessed discriminant validity by comparing this model with others in which the correlation between two constructs was set to one. In each case, the chi-square difference test evinced discrimination. For example, the discriminant test for conflict and current assets was significant ($\chi^2(1) = 14.504$, $p < .05$). Table 3 provides correlations and descriptive statistics for the constructs.

Tests of Hypotheses

The second phase of Anderson and Gerbing's (1988) framework employs structural equation modeling to assess the proposed relationships.² We developed two series of models. The first series assessed the received view, interactive, and parabolic influences of stress. As a means for assessing interaction effects, we incorporated Ping's (1995) estimation technique. Because multiplicative terms raise the possibility of multicollinearity and Type II errors, we mean-centered and standardized the individual measures before developing the nonlinear terms. The initial model examined the linear influences of stress on effectiveness. We employed Lagrange multipliers to assess whether interactive and parabolic influences enhanced the received view model (Bentler 1993). Structural analysis of the linear model indicated

²We also assessed the proposed model using ordinary least squares regression. We extracted the trigonometric and multiplicative terms from equations that regressed them against their relevant linear constructs (see Singh 1998). This procedure eliminates problematic multicollinearity among the antecedent variables. Although the resulting models are biased because of measurement errors, the pattern of results is similar to that resulting from the structural equation models.

some correspondence between the model and data ($\chi^2(529) = 610.049$, $p < .05$; CFI = .976), yet chi-square difference analyses warranted the deletion of paths from the received view model. In addition, sequential chi-square difference tests suggested a group-based communication determinant of cooperative bargaining efforts ($\chi^2(1) = 5.222$, $p < .05$), a parabolic influence of ambiguity on competence ($\chi^2(1) = 5.735$, $p < .05$), and interaction effects for contributions to sales ($\chi^2(1) = 5.763$, $p < .05$). The resulting analysis offers a more parsimonious model ($\chi^2(537) = 605.534$, $p < .05$; CFI = .980).

The second series investigated the G-A-S effects as a competing model to the received view. The G-A-S effects were modeled as sine functions of the ambiguity and conflict variables. This model offers some correspondence to the data ($\chi^2(496) = 607.337$, $p < .05$; CFI = .967), yet chi-square difference tests warrant the deletion of several proposed paths. The Lagrange procedure indicates a relationship between group-based communication and bargaining efforts ($\chi^2(1) = 6.686$, $p < .05$). In addition, personal interaction has a direct influence on contributions to sales ($\chi^2(1) = 7.911$, $p < .05$). The resulting model offers a more efficient representation of the relationships among the constructs ($\chi^2(502) = 607.785$, $p < .05$; CFI = .970). The Friedman comparison test (Rigdon 1999) suggests that this model provides superior fit to the trimmed multiplicative model ($\chi^2(1) = 7.52$, $p < .05$), yet the pattern of results warrants examination of both models (see Table 4).

The influence of role stress on retailer competence is the focus of H_{1a}, H_{1b}, H_{2a}, and H_{2b}. Ambiguity has the anticipated negative, linear effect ($\beta = -.373$, $t = -3.21$), and the parabolic effect ($\beta = .228$, $t = 2.76$) is significant. The graph outlined in Figure 2, Panel A, suggests that these results do not reflect the inverted-U effects outlined in habit formation and activation theory, but they are consistent with the alarm and reactance phases of the G-A-S. The sine function, however, is not significant ($\beta = .155$, $t = 1.82$). The linear ($\beta = .077$, $t = .47$) effects of conflict are nonsignificant. Although the trigonometric ($\beta = -.154$, $t = -1.80$) effects of conflict are marginally significant, they are contrary to the hypothe-

TABLE 3
Correlation Matrix and Descriptive Statistics

	Mean	s	Correlations ^a										
			1	2	3	4	5	6	7	8	9	10	11
1. Current specific assets	5.19	1.02	.65 ^b										
2. Prior specific assets	3.88	2.46	.042	.95									
3. Personal mode	3.14	1.46	.121	.038	— ^c								
4. Group mode	2.99	1.35	.085	.056	.348	— ^c							
5. Written mode	3.45	1.22	.045	-.043	.188	.408	— ^c						
6. Role ambiguity	2.96	1.06	-.030	.263	-.224	-.125	-.132	.81					
7. Role conflict	4.07	1.43	.424	.025	.000	-.047	.012	.280	.84				
8. Competence	4.92	1.23	.042	-.082	.127	-.015	.126	-.299	.075	.69			
9. Customer satisfaction	6.43	.86	.113	.002	.115	.054	.144	-.260	-.027	.234	.88		
10. Contributions to sales	4.67	1.24	.071	-.012	.229	-.025	.047	-.230	.000	.260	.092	.81	
11. Bargaining efforts	3.82	1.49	-.110	-.007	.106	.198	.152	-.401	-.316	.079	.149	.062	.75

^aCorrelations with absolute values greater than .13 are significant at $p < .05$.

^bInternal consistency measures are on the diagonal.

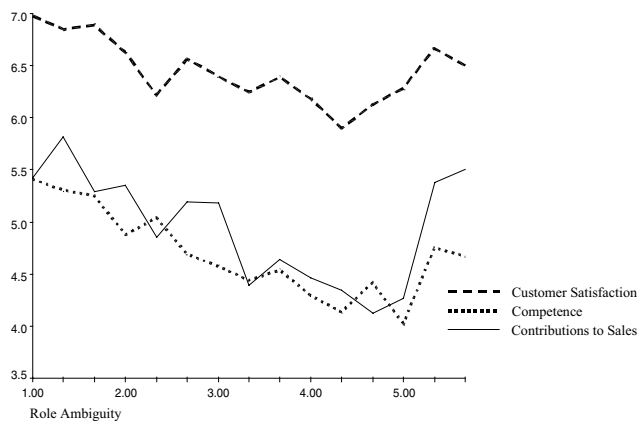
^cFormative scale.

TABLE 4
Structural Equation Models of Antecedents and Consequences of Role Stress

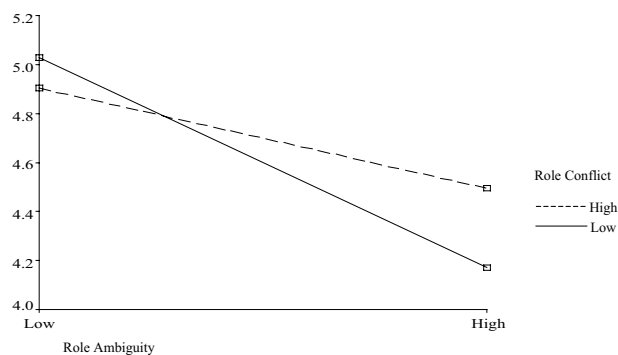
	Initial Received View Model						Trimmed Received View Model					
	Role Ambiguity	Role Conflict	Competence	Customer Satisfaction	Contributions to Sales	Bargaining Efforts	Role Ambiguity	Role Conflict	Competence	Customer Satisfaction	Contributions to Sales	Bargaining Efforts
Prior specific assets	.349 (3.613)	.000 (.002)					.336 (3.675)					
Current specific assets	.079 (.886)	.606 (3.568)						.575 (3.563)				
Personal mode	-.277 (-3.175)	-.086 (-1.230)					-.247 (-3.065)					
Group mode												.192 (2.465)
Role ambiguity			-.406 (-3.055)	-.201 (-1.164)	-.297 (-2.704)	-.164 (-.898)			-.373 (-3.206)		-.292 (-2.939)	
Role conflict			.077 (.473)	.082 (.862)	-.010 (.065)	-.327 (-2.108)						-.311 (1.989)
(Role ambiguity) ²									.228 (2.759)			
Ambiguity × conflict											.186 (2.486)	
Summary Statistics	$\chi^2(529) = 610.049$ CFI = .976 $p < .01$ RMSR = .051						$\chi^2(537) = 605.534$ CFI = .980 $p < .01$ RMSR = .053					
	Initial Trigonometric Model						Trimmed Trigonometric Model					
Role Ambiguity	Role Conflict	Competence	Customer Satisfaction	Contributions to Sales	Bargaining Efforts	Role Ambiguity	Role Conflict	Competence	Customer Satisfaction	Contributions to Sales	Bargaining Efforts	
Prior specific assets	.341 (3.734)	.001 (.011)				.341 (3.798)						
Current specific assets	.020 (.255)	.564 (3.333)					.547 (3.329)					
Personal mode	-.212 (-2.768)	-.060 (-0.872)				-.217 (-2.852)						
Group mode											.205 (2.607)	
Sine (role ambiguity)		.155 (1.815)	.200 (2.550)	.126 (1.945)	.073 (.944)				.157 (1.958)	.165 (2.194)		
Sine (role conflict)		-.174 (-2.023)	-.050 (-.656)	.057 (.914)	.043 (.566)			-.154 (-1.795)				
Summary Statistics	$\chi^2(496) = 607.337$ CFI = .967 $p < .01$ RMSR = .058						$\chi^2(502) = 607.785$ CFI = .970 $p < .01$ RMSR = .056					

FIGURE 2
Effects of Stress on Multiple Facets of Effectiveness

A: Nonlinear Effects of Ambiguity on Effectiveness



B: Interactive Effects of Stress on Contributions to Sales



ses. At low levels, increases in conflict yield greater competence, yet this trend subsides rapidly. Together, these results support H_{1a} , yet they do not support H_{1b} , H_{2a} , and H_{2b} .

H_{3a} , H_{3b} , H_{4a} , and H_{4b} examine the effects of stress on customer satisfaction. The linear ambiguity parameter ($\beta = -.201$, $t = -1.16$) is nonsignificant, yet the trigonometric ($\beta = .157$, $t = 1.96$) function supports H_{3b} . Graphic presentation of the relationship (Figure 2, Panel A) is suggestive of the initial and reactance stages of G-A-S, yet the degradation stage is not observed. H_{4a} and H_{4b} are not supported, given the nonsignificant, linear ($\beta = .082$, $t = .86$) and sinusoidal ($\beta = -.050$, $t = -.66$) effects.

The linear, trigonometric, and interactive influences of stress on contributions to sales are addressed in H_{5a} , H_{5b} , H_{5c} , H_{6a} , and H_{6b} . Ambiguity ($\beta = -.292$, $t = -2.94$) is significantly related to the dependent variable, yet the trigonometric function is also significant ($\beta = .165$, $t = 2.49$). Simultaneous evaluation of the linear and sine functions by means of constraint analysis (Bentler 1993) indicates ($\chi^2(1) = .88$, $p < .35$) that the influence of the linear function

does not exceed that of the sine variable.³ In addition, the linear ($\beta = -.010$, $t = -.06$) and trigonometric ($\beta = .057$, $t = .91$) effects of conflict are nonsignificant. The interactive influence of ambiguity and conflict on contributions to sales is significant ($\beta = .186$, $t = 2.49$), yet the effect is contrary to the hypothesis. To examine these effects, we generate dichotomous levels of ambiguity and conflict on the basis of a median split. We estimate mean levels of contributions to sales for each pair of dichotomous values. The results (see Figure 2, Panel B) indicate that ambiguity has a negative influence on contributions to sales, yet role conflict moderates the effect. When conflict is low, increases in ambiguity lower contributions to sales ($t_{104} = 3.88$, $p < .05$). When conflict is high, increases in ambiguity do not significantly lower contributions to sales ($t_{110} = 1.58$, $p > .05$). Together, these results support H_{5a} and H_{5b} but fail to support H_{5c} , H_{6a} , and H_{6b} .

The influences of stress on bargaining efforts are the focus of H_{7a} , H_{7b} , H_{7c} , H_{8a} , and H_{8b} . In contrast to H_{7a} , H_{7b} , and H_{7c} , ambiguity is not a significant, linear ($\beta = -.164$, $t = -.90$) or trigonometric ($\beta = .073$, $t = .94$) antecedent to bargaining efforts, and the interaction with conflict ($\chi^2(1) = .85$, $p < .36$) does not influence bargaining efforts. H_{8b} is rejected because of the nonsignificant, trigonometric effect ($\beta = .043$, $t = .57$), but the linear effect of conflict outlined in H_{8a} ($\beta = -.311$, $t = -1.99$) is supported.⁴

H_9 – H_{12} examine whether transaction-specific assets influence role stress. Although previous investments in the operating system do not influence conflict ($\gamma = .000$; $t = .00$, $p > .05$), they have a significant influence on ambiguity ($\gamma = .336$; $t = 3.68$, $p < .05$). Currently dedicated assets are associated with conflict ($\gamma = .575$; $t = 3.56$, $p < .05$), yet they do not influence ambiguity ($\gamma = .079$; $t = .89$, $p > .05$). These results support H_9 and H_{12} , but they offer no support for H_{10} and H_{11} .

Facets of communication are the focus of H_{13} and H_{14} . Personal interaction lowers ambiguity ($\gamma = -.247$; $t = -3.07$, $p < .05$), yet it does not influence conflict ($\gamma = -.086$; $t = -1.23$, $p > .05$). The results lend partial support to H_{13} , but not H_{14} .⁵

³The constrained model treating the absolute value of the coefficients as equivalent provides beta coefficients of $-.132$ ($t = -3.08$) and $.205$ ($t = 3.08$) for the linear and sine functions, respectively. In the unconstrained model, the beta coefficient for the linear variable is $-.202$ ($t = -2.20$), and the sine coefficient is $.172$ ($t = 2.30$).

⁴No other parabolic or interactive terms are significant. The Lagrange multipliers do not implicate the squared ambiguity term as an antecedent to customer satisfaction ($\chi^2(1) = 2.600$, $p < .11$), contributions to sales ($\chi^2(1) = .595$, $p < .44$), or bargaining efforts ($\chi^2(1) = .237$, $p < .63$). Similarly, the quadratic conflict term is not related to competence ($\chi^2(1) = 2.182$, $p < .14$), customer satisfaction ($\chi^2(1) = 1.767$, $p < .18$), contributions to sales ($\chi^2(1) = .617$, $p < .43$), or bargaining efforts ($\chi^2(1) = .854$, $p < .36$). The interaction term is unrelated to competence ($\chi^2(1) = 1.897$, $p < .17$), customer satisfaction ($\chi^2(1) = 1.184$, $p < .28$), and bargaining efforts ($\chi^2(1) = .000$, $p < .99$).

⁵The Lagrange multipliers indicate that neither written ($\chi^2(1) = 1.258$, $p < .262$) nor group-based ($\chi^2(1) = .174$, $p < .676$) communication influence role ambiguity. Similarly, the Lagrange multipliers do not indicate an influence of written ($\chi^2(1) = .000$, $p < .99$) or group-based ($\chi^2(1) = 1.091$, $p < .30$) communication on role conflict.

Discussion

Our goal has been to examine role stress in horizontal alliances. The results support the received view prediction of a linear, negative influence of conflict on efforts to streamline bargaining and of ambiguity on competence and contributions to sales. In addition, nonlinear effects that reflect the alarm and reactance phase of G-A-S are observed in the treatment of ambiguity as a determinant of competence, customer satisfaction, and contributions to sales. The data also suggest that conflict moderates the influence of ambiguity on contributions to sales. Role ambiguity is positively associated with dedication to the old operating system, but it is quelled through personal communication. In contrast, conflict is positively associated with current specific investments.

Theoretical Implications

Our analysis of stress in horizontal alliances has incorporated three underlying goals. The first goal has been to examine influences of ambiguity and conflict on facets of effectiveness drawn from the competing values framework (Cameron and Quinn 1999). Human resource development values, such as competence, focus on flexibility and an internal focus, yet market-based values such as contributions to sales favor control along with an external orientation. Retailer efforts to enhance customer satisfaction and other open-system values emphasize autonomy and an external orientation, whereas coordinated bargaining efforts and related factors emphasize an internal, control-based approach to performance. Our treatment of these facets of effectiveness underscores the need to identify the dimensions of effectiveness that are focal to the research. Discrete organizational outcomes not only may be uncorrelated but also may have different antecedents. Theory and practice benefit through explicit identification of the forms of effectiveness addressed in the research. The competing values approach provides a rich theoretical foundation for the analysis of multiple channel outcomes.

Our second objective has been to examine alternative theoretical perspectives on the influences of role stress on organizational outcomes. We offered rationales that are associated with the received view (House and Rizzo 1972), G-A-S (Selye 1950), and the interaction approach (Fried et al. 1998). Although no support is provided to the interaction approach, our research suggests that the form of effectiveness mitigates the merits of the other approaches. The negative effects associated with the received view are observed in the relationship between conflict and bargaining efforts. The received view effects of ambiguity also garner some support, but these effects are augmented by other factors. When the system places value on personal autonomy or an external orientation, there is evidence that managers develop skills that yield competence, customer satisfaction, and contributions to sales despite rising levels of ambiguity. These results suggest that the effects outlined by G-A-S (Selye 1950) are germane when organizational values emphasize personal flexibility or an external focus. These results are tentative,

yet they point to the need to consider multiple theories when examining the influences of ambiguity on effectiveness.

The third objective of our research has been to identify organizational and communicative processes that influence alliances. Dedicated assets have been implicated as determinants of organizational structure, but the deleterious consequences of specific assets have not been considered. Our research indicates that investments in the previous retail operating system raise ambiguity that influences competence, customer satisfaction, and contributions to sales. Current investments in the operating system increase conflict and subsequently constrain bargaining sessions. These results reinforce the need to consider multiple consequences of specific assets. Investments not only have implications for control structures but also influence working relationships.

Our study also has implications for theory addressing the mode of communication employed to distribute information. Mohr, Fisher, and Nevin (1996) emphasize the influence that communication style has on commitment, satisfaction, and coordination, yet scant research has examined the influence of communication media on other facets of interfirm process and performance. Our research suggests that personal modes of communication reduce ambiguity and thus indirectly influence effectiveness. In addition, group-based communication raises the efficiencies of interfirm bargaining. These results are consistent with research that underscores the merits of personal communication (Hallowell 1999). Personal media facilitate dialogue that enables parties to resolve ambiguities, and they facilitate gestalt evaluations of verbal and nonverbal aspects of interaction. The overriding implication of these results is the need to consider the mode of communication. As electronic media evolve and become more personal, research should examine their influence on multiple facets of effectiveness.

Implications for Organizational Design

Although the context, pattern of results, and method limit the extent to which generalizations can be drawn from our research, some tentative recommendations should be acknowledged. First, our work underscores the need to consider the underlying values associated with effectiveness. Prior studies indicate that the influences of stress vary with the form of effectiveness, yet prior research does not offer a framework for categorization and analysis of alternative forms of effectiveness. The competing values framework classifies forms of performance, and our research augments this framework by linking alternative stress theories to the underlying values pursued by the organization. When organizational values downplay personal autonomy, role conflict has a deleterious influence on interfirm outcomes. Therefore, organizations grounded in this perspective should seek decision-making structures that limit the level of stress encountered by trading partners.

Other performance-related values stand in contrast to the control-based internal orientation of a hierarchy, and our research suggests that these values systems have more complex relationships with stressors. Organizational values that emphasize flexibility recognize the merits of enabling local managers to develop their own approaches

to problem solving. Similarly, organizational values that emphasize an external orientation recognize that boundary spanners interact with the environment to exact outcomes. To the extent that these outcomes are valued within the organization, local managers should anticipate some uncertainty. Ambiguities in a role facilitate learning by the role player, and the learning process garners positive outcomes despite increasing levels of stress (Selye 1974).

A second implication of our work is the need to understand that action taken to achieve one objective may have countervailing influences on other factors. For example, training programs designed to make local managers more knowledgeable about the operating system may result in more prolonged bargaining sessions. The competing values framework emphasizes contradictory outcomes realized because of managerial action, yet few research studies have addressed this contingency. Our work reinforces the need for local and area managers to recognize multiple influences of stress.

The results provide insight into mechanisms that can be used to control the level of ambiguity and conflict, notably through transaction-specific investments and personal modes of communication. Our preliminary results suggest that facets of the investment in the retail system yield higher levels of role stress. Consequently, upper-level managers of these networks should recognize that role issues are most likely to emerge among local managers with the most invested in the retail system. Retailers contemplating dual branding should recognize that ambiguity would be greater among local managers with large personal investments in the system. Similarly, the investment made upon entering the new system will raise the level of role conflict. Upper-level management can confront these issues by placing greater emphasis on personal interactions with the local managers. As the perceived level of personal interaction increases, ambiguity diminishes. Impersonal communications offer efficiency in the transmission of policies, but they do not quell conflict or ambiguity. Consequently, organizations that seek to contain role stress should place greater emphasis on personal modes of communication.

Limitations and Further Research

The implications drawn from this study must be viewed in light of the limitations inherent to the research. The outcome variables were based on measures developed in prior research. Replication facilitates comparison, yet replicated measures may provide relatively poor statistical properties. Therefore, the results associated with competence and bargaining efforts are limited by the constraints of two-item scales. The role stress measures have been criticized because of the counterbalancing of positive and negative role stress measures. McGee, Ferguson, and Seers (1989) have developed a procedure for the comparison of the ambiguity and conflict factors with a singular-factor model that incorporates negatively keyed ambiguity items. Their results indicate that the factor structure for Rizzo, House, and Lirtzmann's (1970) measures of ambiguity and conflict is due to the wording of the items. Although our replication is not

fully consistent with their findings,⁶ their study underscores the need to address the breadth of the construct by means of measures that are not confounded by wording biases.

Our Norwegian oil retailing milieu can also be augmented through replication of the model in contrasting empirical contexts. Oil retailing may differ from other contexts on the basis of the resource advantages and near-monopoly power wielded by the refiners. In addition, state participation in the oil industry may limit the ability to generalize our findings to contexts in which government adopts a more laissez-faire approach to commerce. Moreover, the findings associated with retail managers may not replicate with other groups. Retail managers interact with customers, yet they also dedicate time to administrative tasks. The influence of stress on efforts to achieve customer satisfaction and contributions to sales may be more pronounced among customer service employees than among managers.

This survey-based research uses logic drawn from laboratory research to examine the effects of stress on effectiveness. An important distinction between the methods is the manner in which the influences of stimuli are recorded. Laboratory methods chart the within-subjects responses of organisms to progressively higher levels of stress, yet surveys record between-subjects self-reports of multiple levels of stimuli. Future efforts can augment our research by longitudinally tracking responses to stress. Time series provides the opportunity to chart the initial distribution of respondents along the G-A-S thresholds. The researcher has the opportunity to observe whether the G-A-S or other stress-based effects are realized. This design would be particularly insightful in the tracking of stress throughout the planning and completion of an alliance. The research can move beyond the limits of our study and isolate changes in stress incurred through the alliance. Furthermore, the research could track bidirectional changes in the mode of communication and pre-/postmerger specific assets and their influences on stress.

Our research suggests that stress influences organizational outcomes, but it does not consider factors that mitigate the influence of stress. Empowerment should be germane to organizational values that emphasize personal flexibility (Quick et al. 1997). Organizations that empower their managers grant them a level of autonomy that raises perceptions of competence. Because empowered managers operate under a broader set of parameters, they should be equipped to achieve levels of customer satisfaction that exceed those achieved by highly scrutinized managers. If empowerment is to yield desired outcomes, it is essential for

⁶Replication of the procedure developed by McGee, Ferguson, and Seers (1989) failed to illustrate that differences in the scales are solely due to wording or construct discrepancies. The procedure involves the comparison of the oblique ambiguity and conflict factors model with a singular factor model that incorporates negatively keyed ambiguity items. The chi-square for the oblique model is greater than that of the second model ($\chi^2(76) = 210.468$ versus $\chi^2(71) = 205.780$), and the rho statistic for the second model exceeds that of the oblique model ($\rho = .805$ versus $\rho = .801$). In contrast, the delta statistic and CFI for the oblique model ($\delta = .833$, CFI = .805) are greater than the values generated for the second model ($\delta = .821$, CFI = .801). The root mean square residual (RMSR) statistics are identical for the two models (RMSR = .05).

managers to be psychologically involved and interested in the decision-making process (Quick et al. 1997). They must also have the ability to express their thoughts about decisions and believe that their participation is personally relevant to their future. When these conditions hold, empowerment should lead to heightened levels of competence and customer satisfaction.

Analysis of the dominant organizational culture operating within the network can also augment our research on competing values. Multiple facets of effectiveness are associated with specific organizational cultures, yet few firms place equal emphasis on all facets of effectiveness. Singh, Verbeke, and Rhoads (1996) illustrate that organizational archetype mitigates the influences of stress. A procedural archetype characterized by an emphasis on efficiency catalyzes the effects observed in the relationships among ambiguity, conflict, and bargaining efforts, yet an achievement configuration exacerbates the influences of stress on contributions to sales. Research that examines organizational culture and archetype would offer insight into processes by which stress influences effectiveness.

In addition to the institutional constraints on stress, the research can also be augmented by analysis of coping skills.

Coping refers to efforts to manage demands that tax a person's resources (Latack and Havlovic 1992). Research should consider the extent to which the focus of coping is problem-centered and the degree to which the method for coping is cognitive, proactive, and social. When stressors exceed people's ability to cope, burnout is manifested as emotional exhaustion, reduced accomplishment, and depersonalization (Singh, Goolsby, and Rhoads 1994). Stress directly lowers psychological outcomes, yet the eustress effects of stress on behavioral outcomes are cancelled out by the mediating effects of burnout. Research seeking to isolate positive and negative influences of stress would benefit from analysis of coping mechanisms.

In conclusion, the purpose of our research has been to investigate antecedents and consequences of stress in horizontal alliances. Our results suggest that economic conditions and communicative mechanisms influence role stress. Furthermore, ambiguity and conflict influence multiple facets of effectiveness, and the form of influence is mitigated by the underlying values pursued by the organization. We hope that these findings are informative to retail management and interorganizational theory.

APPENDIX

Measurement Models

Scale Items	Initial Model ^a	Trimmed Model ^a
Investment in Current System		
It is vital that a manager builds strong relationships within our company.	.322 ^b	.324 ^b
The refiner's procedures and policies are complicated.	.807	.810
It is difficult for a manager to learn all tasks necessary to function effectively in the system.	.655	.653
A newcomer to the system must learn our language and culture before being capable to function effectively.	.356	.352
Investment in Previous System		
Before the merger, it was difficult for a new station manager to work effectively.	.869 ^b	.869 ^b
The refiner's procedures and policies were complicated prior to the merger.	.917	.917
Before the merger, it was difficult for a manager to learn all tasks necessary to function effectively in the system.	.951	.951
Prior to the merger, a newcomer to the system had to learn the company's language and culture before being capable to function effectively.	.901	.901
Role Ambiguity		
I feel certain about how much authority I have. (r)	.422 ^b	.422 ^b
I know what my responsibilities are. (r)	.616	.615
I have just the right amount of work to do. (r)	.422	.420
I know that I have divided my time properly. (r)	.566	.567
I know exactly what is expected of me. (r)	.751	.752
Explanation of what has to be done is clear. (r)	.738	.738
I perform work that suits my values. (r)	.594	.594
Role Conflict		
I receive assignments without the personnel to complete the task.	.818 ^b	.822 ^b
I have to circumvent rules or policies to complete assignments.	.635	.644
I receive incompatible requests from two or more refiner personnel.	.446	.446
I often get assignments without adequate resources and materials to execute them.	.843	.841
I work on unnecessary things for the refiner.	.709	.703
I have to work under vague directives or orders.	.389	—
Competence		
We believe that we have acquired the necessary business skills to operate successfully in my market.	.721 ^b	.724 ^b
We believe that we have a great deal of knowledge about the features and attributes of the refiner's products.	.731	.728

APPENDIX

Measurement Models

Scale Items	Initial Model ^a	Trimmed Model ^a
Customer Satisfaction		
We do everything we can to make our customers happy.	.715 ^b	.715 ^b
We provide quality assistance in the solution of any problem involving the refiner's products.	.902	.901
We provide information that lowers uncertainty about use of the refiner's products.	.705	.705
Contributions to Sales		
Relative to other dealers, we have been successful in selling the refiner's products.	.820 ^b	.820 ^b
Compared to other dealers, we have achieved a higher level of market penetration for the refiner.	.946	.946
We have contributed more to the refiner's business in this market than other dealers.	.746	.746
Bargaining Efforts		
Our meetings with the refiner's representatives are very effective and systematic.	.753 ^b	.751 ^b
Both parties are always well prepared in the meetings with the refiner so that decisions can be made.	.804	.805
Summary Statistics		
χ^2	635.546	589.979
(d.f.)	406	377
p-Value	.001	.001
RMSR	.050	.047
CFI	.921	.926

^aAll factor loadings have t-values that exceed 2.0.

^bThese items are fixed for the purpose of scaling.

Notes: Items marked (r) were reverse scored.

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